H-B-181c
March 21, 1968
SUPERSEDING
Fed. Spec. H-B-181b
October 2, 1963

FEDERAL SPECIFICATION

BRUSH, CHASSIS AND RUNNING GEAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type and size of chassis and running-gear cleaning brush (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951 - Synthetic Filaments for Brushes; General Specification for.
PPP-B-636 - Box, Fiberboard.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Office in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-W-6110 - Wood, Determination of Moisture Content of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

American Trucking Association, Inc.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P. Street, N.W., Washington, D. C. 20036).

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample brush covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. The brushes delivered under this specification shall be the manufacturer's standard commercial product except for any changes necessary to comply with specification requirements.

3.3 Materials. Materials shall conform to the applicable specifications and requirements specified hereinafter. Materials not specifically defined shall be of the quality normally used for the purpose in commercial practice.

3.3.1 Polypropylene. The polypropylene shall be 0.030 inches in diameter and shall conform to class P, type II of H-S-951.

3.3.2 Palmetto. The palmetto fiber shall be oil treated in accordance with commercial practice.

3.3.3 Wood. The block shall be hardwood, of a dense, close grained species such as beech, birch, sycamore, or maple. The wood prior to machining shall be thoroughly and uniformly dried to a moisture content not exceeding 12 percent when tested as specified in 4.4.1.1.

3.3.4 Staple wire. Staple wire shall be not less than 0.046-inch in diameter, commercial steel staple wire, corrosion-resistant or treated to resist corrosion in accordance with manufacturer's commercial practice.

3.4 Construction. The brush shall conform to the requirements specified herein and in table I, and shall be similar to figure 1. Figure 1, incorporated in this specification, is included for identification only and is not intended to preclude the purchase of brushes otherwise in accordance with this specification.

3.4.1 Filler material. The filler material shall be either polypropylene specified in 3.3.1 or palmetto fiber specified in 3.3.2, at the option of the supplier.

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TABLE I. Minimum physical requirements

Block dimensions, inches:	
Length:	
Overall	19-1/2
Brush part	3-1/8
Width:	
Handle (end)	3/4
Brush part	2-1/2
Thickness:	7/8
Holes:	
Diameter	9/32
Number	56
Angle (at sides and end)	30°
Filler material, as applicable:	
Weight, ounces 1/	
Polypropylene	2-3/8
Oiled palmetto	3-7/8
Length clear of block, inches	2

1/ When tested as specified in 4.5.1, as applicable.

3.4.2 Block. The block shall be fabricated from a single piece of the wood specified in 3.3.3. Filler holes perimeters shall be spaced not less than 1/6 inch to an adjacent hole. The edges of the entire block shall be evenly rounded. A 3/16 diameter hole shall be centered 1-1/4 ± 1/4 inches from the end of the handle.

3.4.3 Set. The filler material shall be cut to double length and folded in the center and stapled into the holes of the block. The staples shall be made of the wire specified in 3.3.4. The tufts shall not loosen, nor pull out of the brush when tested as specified in 4.5.2.

3.5 Finish. The blocks shall be sanded or tumbled smooth and at the option of the manufacturer, the blocks shall be coated either with a clear organic coating or tumbled waxed in accordance with the manufacturer's commercial practice.

3.6 Marking for identification. The supplier shall permanently and legibly indent his name, or trademark of such known character that it is easily identifiable with said manufacturer, on the handle of each brush.

3.7 Workmanship. Blocks shall be free of loose grain, splinters and rough edges. Staples shall not be exposed. Filler material shall be clean, evenly trimmed and shall be free of foreign matter. Brushes shall be free from defects which may affect their durability, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirement.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects listed in table II, specified dimensions and tested as specified in 4.5.1.1 or 4.5.1.2 as applicable and 4.5.2.

4.3 Sampling. Sampling shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.4 Inspection.

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4.4.1 Component and material inspection. In accordance with 4.1, component and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Moisture content of block. Test shall be performed on each lot of wood offered for inspection prior to machining for moisture content in accordance with any method of MIL-W-6110, to determine compliance with 3.3.3. The sample unit shall be one piece of cut wood. Two determinations shall be made on each sample unit and the result shall be the average of the two determinations. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 defects expressed in terms of defects per hundred units.

4.4.2 End item inspection. A lot shall consist of all brushes offered for inspection at one time. The sample unit for this inspection shall be one complete brush.

4.4.2.1 Visual examination. Examination shall be made of the brush for defects set forth in table II. The inspection level shall be level I with an (AQL) of 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE II. Visual examination

Examine	Defect
Finish	Not sanded or tumbled smooth
	Not coated with either a clear organic coating or wax
Block	Not single piece of wood; number of holes for filler material less than specified
	Edges not rounded; handle not provided with a hole; loose grain; splinter or rough edge.
Staple	Exposed
Filler material	Not clean; not evenly trimmed; evidence of foreign matter
	Palmetto fiber not oil treated
Marking	Identification marking omitted, incomplete, or illegible

4.4.2.2 Dimensional examination. Examination shall be made of the brush for defects in dimensions. Any dimension not within specification requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.4.2.3 Testing of the end item. Each sample brush shall be tested in accordance with 4.5.1.1 or 4.5.1.2 as applicable and 4.5.2. The inspection level shall be S-2 with an AQL of 2.5 defects expressed in terms of defects per hundred units.

4.4.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE III. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing or damaged

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4.4.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.5 Tests.

4.5.1 Weight of filler material. The filler material shall be removed from the brush prior to weighing.

4.5.1.1 Polypropylene. Unconditioned polypropylene shall be weighed to determine compliance with table I.

4.5.1.2 Palmetto fiber. The palmetto fiber shall be conditioned for not less than 16 hours in air at a relative humidity of 50 $\pm$ 4 percent, and at a temperature of 73° $\pm$ 2°F. The conditioned fiber shall then be weighed in the same atmosphere to determine compliance with table I.

4.5.2 Set retention. Grasp the tuft with a suitable holding device and apply a force by using a scale or weight of not less than 10 pounds in the direction tending to pull the tuft from the block to determine compliance with 3.4.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Twelve brushes shall be packaged in a snug-fitting fiberboard box conforming to style RSC type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Brushes shall be packaged six wide and two deep, each brush reversed (head to handle) and a sheet of rigid paperboard or fiberboard inserted between the two layers of brushes. Each box shall be closed by covering all seams full length with strips of 2 inch minimum width tape conforming to PPP-T-76 or type III, grade B of PPP-T-45. The center strip shall extend a minimum of 2 inches onto the end panels. Alternatively, the bottom flaps may be sealed with adhesive or by metal stitching.

5.1.2 Level C. Brushes shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Sixty brushes, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Twelve brushes, packaged as specified in 5.1.1, shall require no overpacking. Alternatively, 60 brushes, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.3 Level C. Brushes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification Rules, or National Motor Freight Classification Rules, or Parcel Post Regulations, when applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The brush covered by this specification is intended to be used in cleaning of vehicular chassis and running gear.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) When preproduction sample is required (see 3.1)
(c) Selection of applicable levels of packaging and packing

6.3 Standard packaging (civil agencies). The standard packaging and packing for civil agencies shall be as specified in 5.1.1 and 5.2.2, respectively.

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - GL
Navy - YD
Air Force - 84, 85

User activity:

Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

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GSA
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INT
JUS
JUS-FPI
VA

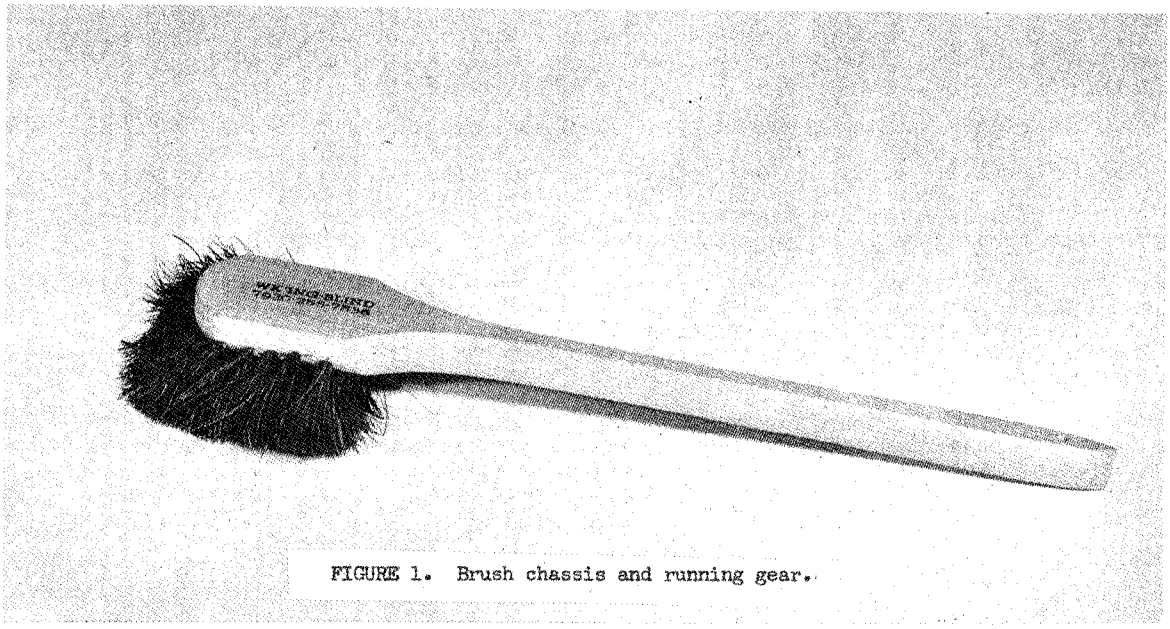


FIGURE 1. Brush chassis and running gear.

Orders for this publication are to be placed with General Services Administration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Brush, Chassis and Running Gear - H-B-181c

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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